

2016.0 RANGE ROVER (LG), 501-20

SEATBELT SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the safety belt system and operation, refer to the relevant Description and Operation section of the workshop manual.
REFER to: Safety Belt System (501-20 Safety Belt System, Description and Operation).

SAFETY INFORMATION

WARNINGS:

- To avoid accidental deployment the back-up power supply must be depleted before beginning any work on the SRS system or its components. Failure to follow this instruction may result in personal injury.
- Do not use a multimeter to probe an SRS module. It is possible for the power from the meter battery to trigger the activation of the module. Failure to follow this instruction may result in personal injury.

NOTE:

It is advisable not to use a cellular phone or to have a cellular phone in close proximity when working on the SRS system or components.

Power supply depletion

Before beginning any work on the SRS system or related components:

1. Remove the ignition key.
2. Disconnect the battery leads, ground first.
3. Wait 2 minutes for the power circuit to discharge.

There are comprehensive instructions on the correct procedures for SRS system repairs in the manual. Refer to the relevant section of the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

- 1. Verify the customer concern by operating the safety belt.
- 2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Check for the installation of non-standard accessories which may affect or obstruct the function of the safety belt system ▪ Frayed or damaged webbing ▪ Buckles/Stalks, etc ▪ Reels ▪ Retractors ▪ Sensor(s) ▪ Pretensioner(s)	▪ Fuses ▪ Wiring harness ▪ Warning lamp bulb(s) ▪ Impact sensor(s) ▪ Buckle sensor(s) ▪ Pretensioner(s) ▪ Restraints control module (RCM)

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
- 4. If the cause is not visually evident, carry out the test methods described below, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the relevant DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Seatbelt - Jammed	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A .

Seatbelt - Not retracting	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A.
Seatbelt - Not working	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A.
Seatbelt - Slow retraction	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A.
Seatbelt - Binding	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A.
Seatbelt - Inoperative	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A.
Seatbelt - Intermittent	▪ (ALR) automatic locking retractor in use (normal operation)	GO to Pinpoint Test A.

	▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	
Seatbelt - Jamming	▪ (ALR) automatic locking retractor in use (normal operation) ▪ Twist in webbing ▪ Webbing guide incorrect location ▪ Foreign object in webbing route ▪ Safety belt reel is incorrectly installed	GO to Pinpoint Test A .

COMPONENT TESTS

POOR RETRACTION

If a safety belt does not retract correctly, check that the anchor covers and trim bezels are correctly installed and not rubbing against the safety belt webbing. Where necessary, check the safety belt webbing is not rubbing at one end of the retractor cover slot and if so, correct by loosening the retaining bolt, aligning the retractor to centralize the safety belt webbing and re-tighten the bolt.

The vehicle is equipped with two front and three rear inertia reel safety belts. These safety belts are "dual sensitive" which means that they have:

- a vehicle motion sensor, which locks the safety belt webbing under braking, cornering, on steep hills and in adverse camber conditions.
- a webbing motion sensor, which locks when the safety belt webbing is quickly extracted.

Both systems should be fully operational and can be checked by the tests below:

DYNAMIC TESTS

Either of the following two procedures may be used to check the correct operation of the safety belts. Both methods require two people but note that people of larger than average size should not be asked to conduct these tests. This is to avoid the possibility of a fully unrolled safety belt webbing being mistaken for a correctly locked safety belt retractor.

TEST METHOD 1 (BRAKING)

WARNING:

It is important that during this test, the wearer allows the safety belt to provide the restraint, the wearer should not attempt to anticipate the sudden deceleration and the driver should not brace themselves against the steering wheel. However, both the driver and the passenger must prepare themselves for the possibility that the safety belt will not lock. The passenger should hold their hands in front of them, just clear of the instrument panel or front seat backrest, depending on which belt is being tested. Failure to follow these instructions may result in personal injury.

- Select a quiet or private stretch of road. Make sure that the road is clear and that full visibility is maintained at all times.
- Both driver and passenger should adopt a normal, comfortable seating position. Both occupants should wear the safety belts and the safety belt webbing should be correctly adjusted.
- Proceed at a speed of approximately 10 km/h (6 miles/hour).
 - Do not exceed 10 km/h (6 miles/hour) for this test
- Apply the foot brake sharply to stop the vehicle. If the vehicle motion sensitive lock mechanism is operating correctly, the safety belt webbing will lock and restrain the wearer.
- Conduct the test twice in each front and rear passenger seat position.

- Any safety belt retractor which does not restrain the wearer during this test must not be reused. A new safety belt should be installed.

TEST METHOD 2 (TURNING CIRCLE)

This method requires a flat open area of private road, sufficient for the vehicle to be driven in a continuous circle on full lock.

- The driver should wear the safety belt provided and the belt webbing should be correctly adjusted.
- The passenger should occupy a rear seat with the safety belt correctly adjusted.
- Start the engine and, with the steering on full right-hand lock, drive the vehicle in a continuous circle at 16 km/h (10 miles/hour).
 - Do not exceed 16 km/h (10 miles/hour) for this test
- When the speed is stable, the passenger should attempt to slowly extract the safety belt webbing from each safety belt retractor in turn. If the vehicle motion sensitive lock mechanism is operating correctly, it will not be possible to extract the webbing.
- Any safety belt retractor from which it is possible to extract the webbing during this test must not be used. A new safety belt should be installed.

SAFETY BELT WEBBING SENSOR TEST

With the vehicle stationary and on level ground take firm hold of the safety belt webbing (on the tongue side of the upper safety belt anchor) and pull out quickly. The retractor should lock within 0.25m (10 inches), preventing further webbing release. Any safety belt retractor from which it is possible to extract further webbing must not be used. A new safety belt should be installed.

D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

PINPOINT TEST

NOTE:

During normal conditions the (ALR) automatic locking retractor can operate giving the impression of a jammed safety belt this can be overcome by drawing 20mm of webbing from the belt reel with moderate force, the belt reel should then operate correctly. Check the operation, if the webbing moves freely then retracts correctly no further action required, do not replace safety belt.

NOTE:

The safety belt reel has a guide installed that can cause the safety belt reel to operate poorly if incorrectly located.

NOTE:

Incorrect installation / location of the safety belt reel can affect operation.

PINPOINT TEST A : DIAGNOSTIC GUIDE FOR THE SECOND AND THIRD ROW SAFETY BELTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: SAFETY BELT	
	1 Visually inspect the condition of the safety belt.
	Is the safety belt fully retracted? Yes The safety belt is fully retracted, no slack in the webbing. GO to A2. No Not fully retracted, the webbing is slack. GO to A3.

A2: REQUIRED FORCE

	1 Draw the webbing from the safety belt reel with moderate force then release the webbing. Carry out this operation twice.
	Does the webbing move freely then retract correctly? Yes No further action required, do not replace safety belt. No GO to A3.

A3: WEBBING SLACK - NOT FULLY RETRACTED

	1 Remove any trim panels to expose the D loop (anchor point) and safety belt reel. Check the webbing is not trapped or twisted.
	Does the webbing move freely then retract correctly? Yes While refitting the trim panels, ensure there are no obstruction and the webbing does not catch or rub. Check the operation, if the webbing moves freely then retracts correctly no further action required, do not replace safety belt. No GO to A4.

A4: TRAPPED OR TWISTED SAFETY BELT WEBBING

	1 If the webbing is twisted due to a fold passing through the anchor point (do not remove the safety belt reel) feed the twist back through the anchor point to remove. If the webbing is trapped, release and inspect for damage, if the webbing is damaged replace the safety belt.
	Does the webbing move freely then retract correctly? Yes If the webbing moves freely then retracts correctly no further action required, do not replace safety belt. While refitting the trim panels, ensure there are no obstruction and the webbing does not catch or rub. No GO to A5.

A5: SAFETY BELT WEBBING GUIDE LOCATION

	1 Visually and manually inspect the safety belt reel webbing guide for location and installation.
	Is the safety belt reel webbing guide for location and installation correct and secure? Yes GO to A6. No Refit the safety belt reel webbing guide ensure the fixing lug is properly located in the retractor frame. When correct draw the full length of the webbing from the reel then allow the belt to retract, the webbing will now be central on the spindle. Check the operation, if the webbing

	moves freely then retracts correctly, no further action required, do not replace safety belt. While replacing the trim panels, ensure there are no obstruction and the webbing does not catch or rub.
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A6: SAFETY BELT REEL LOCATION	
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	1 Confirm that the safety belt reel is correctly installed, check for correct location of both anti rotational pins if present.
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	Is the safety belt reel is correctly installed?
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	Yes
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	Check the operation, if the webbing moves freely then retracts correctly, no further action required, do not replace safety belt. While replacing the trim panels, ensure there are no obstruction and the webbing does not catch or rub.
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	Correctly install the safety belt reel ensuring correct location of both anti rotational pins if present. Check the operation, if the webbing moves freely then retracts correctly, no further action required, do not replace safety belt. While replacing the trim panels, ensure there are no obstruction and the webbing does not catch or rub.
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